

Classified Effects of Nanofillers on DC Breakdown and Partial Discharge Resistance of Polypropylene/Alumina Nanocomposites

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ABSTRACT

The enhancement of DC breakdown strength in polypropylene/alumina nanocomposites (NCs) with low filler content (1 wt%) while the obviously increased partial discharge (PD) resistance at higher filler content (15 and 20 wt%), are observed. Possible mechanisms underlying such contrastive performance are proposed based on considering a combination of electric conductivity, traps distribution and morphology of the eroded test specimen. The DC breakdown is closely related to trap-controlled charge transport and the property is improved for NCs with low filler content. Comparatively, the PD resistance of NCs with higher filler content exhibits longer failure time and more spreading of the eroded area on specimen surface. A model of PD resistance of NCs is proposed on the basis of synergistic effects of energy dissipation by nanoparticle scattering and charge dissipation. The scattering effect on charged ions induced by nanoparticles plays an important role in energy dissipation, and faster charge dissipation along the sample surface induced by shallower traps mitigates energy concentration. Thus two aspects contribute to the improved PD resistance. It is concluded that the effects of nanofillers on short-term DC breakdown and long-term PD resistance are different. A low filler content improves short-term DC breakdown performance whereas a high filler content enhances long-term PD resistance.

Index Terms — breakdown, PD resistance, interface, traps, energy dissipation, scattering

1 INTRODUCTION

POLYPROPYLENE (PP) is being widely used in power capacitors and also considered as a potential electrical insulation material for high voltage DC cables owing to its high breakdown strength and excellent space charge suppression [1,2]. The breakdown (short-term breakdown and long-term failure) of PP insulation under high electric field is the main consideration for the safe operation of capacitors and HVDC cables. The development of nanocomposites (NCs) provides potential solution for making insulating materials with higher voltage endurance and lifetime [3].

A number of studies showed that nanofillers could significantly change the short-term breakdown and long-term failure performance of NCs [1–14]. Due to the large specific

surface area of nanoparticles, substantial interface regions are formed between the nanofillers and the polymeric matrix [3]. It was considered that the interface and its physical property play a crucial role on the dielectric performance of NCs [3–6]. For the short-term breakdown, the scattering or attracting effect of electrons by Coulomb force [3–5] at the interface was considered responsible for improving dielectric breakdown strength. Besides, the significant role of trap characteristics induced by the interface on charge transport dynamics and thus the breakdown performance was proposed [5–9]. In the case of long-term failure, the effect of nano-segmentation with the interaction zone [10] was considered a mechanism for the improved partial discharge (PD) resistance. Some other investigations reported that the better PD resistance could be originated from the blocking role of layered silicate fillers [11], the weakened electric field induced by high dielectric permittivity [12, 13], the shorter inter-particle distances achieved by uniform dispersion [14].

These findings facilitate a better understanding of the short-term breakdown and long-term PD failure of NCs. The indirect effect of the interfacial traps, induced by nanofillers, on the dielectric breakdown and PD resistance had been investigated in some studies. The influence of nanofillers on the dielectric breakdown can be understood through charge transport determined by traps. However, the change of eroded surface morphology induced by PD and direct effect of nanoparticles on the enhanced PD resistance remain unclear. The mechanism for the increase of long-term surface PD resistance as a function of nanofiller contents needs to be further studied.

In this paper, PP/Al₂O₃ NCs samples with different filler contents were fabricated and put to tests for DC breakdown and PD resistance experiments. Combining electric conductivity and trap characteristics, the DC breakdown related to the trap-controlled charge transport process was analyzed. In addition, the eroded surface morphologies after PD were obtained. The indirect effect of interfacial traps and the direct effect of surface scattering by nanofillers are discussed in detail. It was concluded that the effects of nanofillers on the mechanism for short-term DC breakdown and long-term PD resistance are different. The findings drawn from this work provide design guidance on using appropriate nanofiller contents to achieve desired short-term and long-term electrical performance.

2 SAMPLES AND EXPERIMENTS

2.1 SPECIMEN PREPARATION

Isotactic PP pellets (HC312BF, Borealis) were used for the matrix. Alumina nanoparticles (Al₂O₃, Beijing D.K. Nano Technology) with diameter of 50 nm were used as the fillers. To improve the dispersion of nanofillers, surface modification was applied to Al₂O₃ nanoparticles by silane coupling agent KH570 (3-Methacryloxypropyltrimethoxysilane). The PP matrix was melt-blended with the nanoparticles in a HAAKE PolyLab QC torque rheometer at a rotational speed of 150 rpm for 30 min at 453 K. Then, pure PP and its NCs with different weight percentages (1, 7 wt%) of nanofillers were obtained. NCs with higher contents (15, 20 wt%) were also prepared to highlight the comparison of DC breakdown and PD resistance. For referencing, the NC with 1wt% is referred to as PP/Al₂O₃-1wt% and similar labeling applied to other NCs. Thin film samples with thicknesses of 100 μ m and 150 μ m were hot-pressed by using a compression molding press under 20 MPa for 10 min at 453 K.

2.2 SPECIMEN MORPHOLOGY

A KEYENCE VE-9800S scanning electron microscope (SEM) was employed for characterization of the cross section of the samples. The morphology and dispersion of nanoparticles in PP matrix were examined. In addition, the eroded surface morphologies of pure PP and its NCs after PD erosion were also studied by SEM.

2.3 DIELECTRIC PERFORMANCE

Dielectric spectrum measurements were conducted on the 150 μ m PP samples by using a broadband dielectric spectrometer (Concept80, Novocontrol Technology). AC voltage of 1 V_{rms} and frequency varying from 10⁻¹ to 10⁵ Hz was applied to samples at 298 K.

Field-induced conductivities of the 150 μ m PP samples were measured by a three-electrode system with a Keithley 6517B electrometer. The stable current flowed through the test sample was recorded after applying a fixed DC voltage for 5 min. A series of current measurements was done under different applied voltage. After each voltage application, the sample was short-circuited to remove the charges and then the next measurement with a higher applied voltage was performed. The curves of current versus voltage for each sample were obtained. The current density $j(E)$ can be calculated by knowing the upper electrode diameter (25 mm) and thus the $j(E)$ - E curves can be obtained.

Thermally stimulated depolarization current (TSDC) measurements were performed on 150 μ m PP samples by using a Novocontrol TSDC apparatus. Electrodes with radii of 15 mm were used. The TSDC procedure was detailed in previous work [13] and adapted in this study, as shown in Figure 1. The samples were polarized under 250 V at 383 K for 20 min. After polarization, the samples were cooled down to 248 K at a cooling rate of 30 Kmin⁻¹. On reaching this temperature, the samples were short-circuited for 5 min and then heated at a rate of 3 Kmin⁻¹. The depolarization current was measured during this heating process until 423 K.

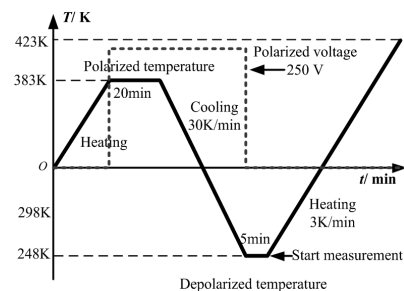


Figure 1. TSDC experimental procedure.

Short-term DC breakdown experiments on the 150 μ m samples were conducted in transformer oil at 298 K. The test specimen was sandwiched between two spherical copper electrodes of 25 mm diameter, as shown in Figure 2a. DC voltage with a linearly ramping rate of 1 kV/s was applied to the specimen until breakdown.

Long-term PD resistance experiments were conducted on the 100 μ m PP film samples with a fixed AC voltage of 7 kV. A needle-plate electrode setup was used as shown in Figure 2b. The gap distance between the needle tip (radius of curvature with 11.75 μ m) and the surface of the test specimen was 1 mm. During high voltage application, the specimen surface was exposed to the discharges from the needle electrode. The time until breakdown occurred was recorded which was in the order of several thousand seconds.

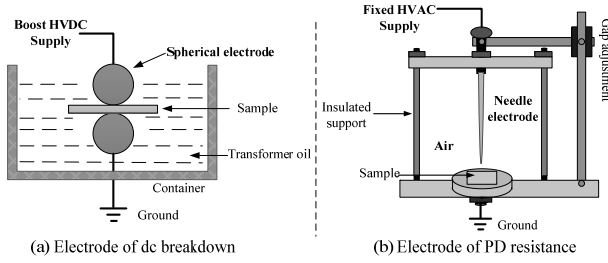


Figure 2. Electrode schematics for DC breakdown and PD resistance tests

3 EXPERIMENTAL RESULTS

Figure 3 shows the SEM images of PP samples. The morphology of NCs indicates a large difference compared with that of pure PP. Numerous spots can be observed in PP/Al₂O₃ NCs and the distribution of nanoparticles in the PP matrix is relatively uniform. There is little agglomeration in those NCs with high nanofiller contents.

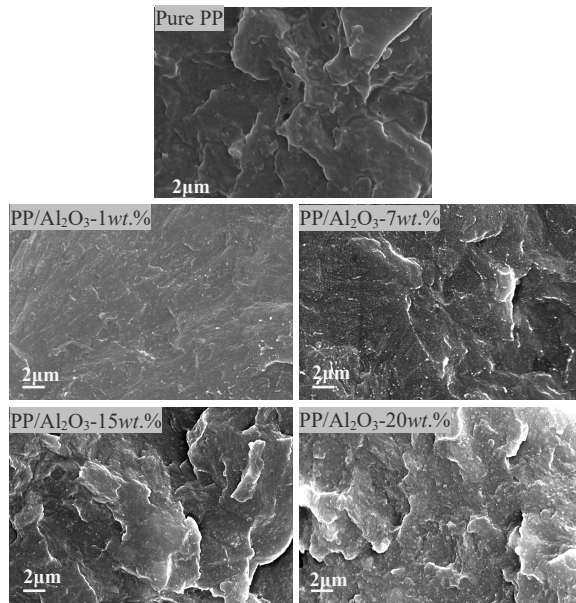


Figure 3. SEM images of PP samples with different filler contents.

3.1 DIELECTRIC PERMITTIVITY

The real permittivity ϵ_r and imaginary permittivity ϵ'' of the PP samples are shown in Figure 4a and Figure 4b, respectively. The values of ϵ_r for PP/Al₂O₃ NCs are increased compared with that of pure PP. With increasing nanofiller

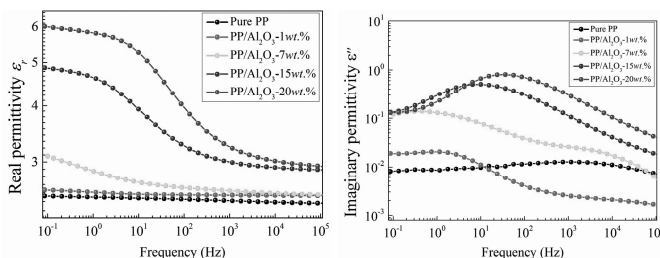


Figure 4. Real permittivity and imaginary permittivity as a function of frequency for PP samples.

content, ϵ_r increases. A clear enhancement of real permittivity at low frequency (10^{-1} – 10^0) is observed for NCs with high contents (>7 wt%). The big permittivity of Al₂O₃ nanoparticles (≈ 9.4) and the interfacial polarization between the nanofillers and polymer matrix could increase the dielectric permittivity of NCs [2]. For 15 and 20 wt% NCs, a substantial step occurs over the frequency range 10^0 – 10^2 Hz, and two distinct relaxation peaks appear in this frequency range, which could be induced by the numerous interfacial polarization.

3.2 FIELD INDUCED CONDUCTIVITY

Figure 5 shows the relationship between the current density and the external dc electric field. The PP/Al₂O₃-1wt% sample consistently exhibits the lowest current density irrespective of the applied electric field, successively followed by pure PP, PP/Al₂O₃-7, 15, and 20 wt%. According to [15, 19] regarding the effect of applied electric field on the electrical conduction in dielectrics, the $j(E)$ – E characteristics obey Ohmic law in the low-electric-field region and the trap-related space charge limited conduction (SCLC) [15] in the high-electric-field region:

$$j(E) = \begin{cases} \sigma E, & \text{Ohmic region} \\ \frac{9\epsilon_0\epsilon_r\mu_{eff}E^2}{8d}, & \text{SCLC region} \end{cases} \quad (1)$$

where $j(E)$ is the current density in Am^{-2} ; σ is the conductivity in Sm^{-1} ; E is the external electric field in Vm^{-1} ; μ_{eff} is the trap-controlled effective mobility with considering the traps, in $\text{m}^2\text{V}^{-1}\text{s}^{-1}$; and d is the sample thickness in m. The dielectric permittivity ϵ_r at 10^{-1} Hz in Section 3.1 is adopted in the calculation of carrier mobility.

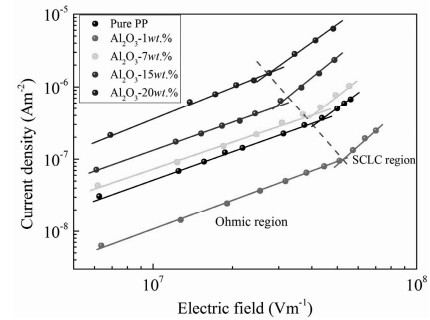


Figure 5. $j(E)$ – E characteristics of pure PP and its NCs.

The average conductivity σ and the effective mobility μ_{eff} are calculated by Equation (1) and tabulated in Table 1. The results show that σ and μ_{eff} of PP/Al₂O₃-1wt% are reduced as compared to pure PP. On the contrary, those of PP/Al₂O₃-7 wt%, 15 wt% and 20 wt% are successively increased.

3.3 TRAP PARAMETERS

TSDC results together with the fitted curves are shown in Figure 6. The PP/Al₂O₃-1wt% exhibits a shift in the peak temperature T_p to a higher temperature and increase in intensity compared with that of pure PP. For PP/Al₂O₃-7, 15, and 20 wt%, T_p successively shifts to a lower temperature and increases in intensity compared with that of pure PP. The TSDC

Table 1. Calculated conductivity and effective carrier mobility.

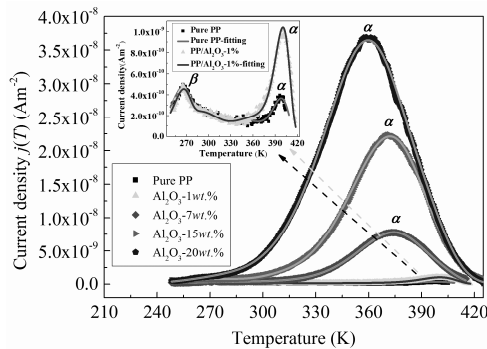
Specimen	Low field σ (Sm ⁻¹)	High field μ_{eff} (m ² V ⁻¹ s ⁻¹)
Pure PP	6.35×10 ⁻¹⁵	1.29×10 ⁻¹⁵
PP/Al ₂ O ₃ -1wt%	1.41×10 ⁻¹⁵	2.66×10 ⁻¹⁶
PP/Al ₂ O ₃ -7wt%	8.79×10 ⁻¹⁵	1.62×10 ⁻¹⁵
PP/Al ₂ O ₃ -15wt%	1.49×10 ⁻¹⁴	2.74×10 ⁻¹⁵
PP/Al ₂ O ₃ -20wt%	4.48×10 ⁻¹⁴	5.87×10 ⁻¹⁵

results are analyzed by the TSDC Equation (2), as shown in [16]. The trap energy E_a is obtained by fitting the $j(T)$ - T curves [17] with Equation (2). The quantity of trapped charges Q_{TSDC} is the integration of $j(T)$ - T curves, as expressed in Equation (3) [16]:

$$j(T) = A \exp \left[-\frac{E_a}{k_B T} - \frac{B}{\beta_T} \int_{T_0}^T \exp \left(-\frac{E_a}{k_B T} \right) dT \right] \quad (2)$$

$$Q_{TSDC} = \frac{60S}{\beta_T} \int_{T_0}^T j(T) dT \quad (3)$$

where $j(T)$ is the depolarization current density in Am⁻²; A and B are constants shown in [16]; β_T is the heating rate in Ks⁻¹; k_B is the Boltzmann constant; T is the temperature in K; and S is the electrode area in m².

**Figure 6.** TSDC results and corresponding fitted curves.

The trap parameters are listed in Table 2. Compared with pure PP, the deep trap energy (peak α) of PP/Al₂O₃-1wt% is increased from 1.17 to 1.34 eV, while that of PP/Al₂O₃-7, 15 and 20 wt% are obviously decreased down to 0.53 eV. The shallower traps are obtained in those samples with higher nanofiller content. In addition, the trapped charges Q_{TSDC} of peak α increase with increasing nanofiller content. It should be noted that, the peak β can be clearly seen for pure PP and PP/

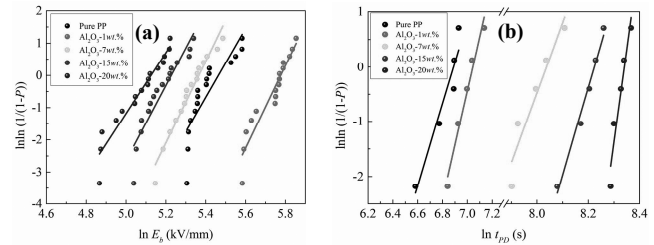
Table 2. The calculated trap parameters.

Specimen	Peak	T_p (K)	E_a (eV)	Q_{TSDC} (nC)
Pure PP	α	397	1.17	0.15
	β	263	0.45	0.19
PP/Al ₂ O ₃ -1wt%	α	401	1.34	0.41
	β	265	0.46	0.17
PP/Al ₂ O ₃ -7wt%	α	376	0.82	5.58
PP/Al ₂ O ₃ -15wt%	α	371	0.6	17.4
PP/Al ₂ O ₃ -20wt%	α	359	0.53	32.9

Al₂O₃-1wt% whereas for high nanofiller contents of 7, 15 and 20 wt% NCs, the peak β is covered up.

3.4 DC BREAKDOWN AND SURFACE PD RESISTANCE PERFORMANCE

Two-parameter Weibull distribution was employed for failure statistics, the details can be found in [18]. Weibull distribution of DC breakdown strength and PD resistance time are shown in Figure 7a and Figure 7b, respectively.

**Figure 7.** Weibull distribution of (a) dc breakdown, and (b) PD resistance.

The calculated scale parameter α , and the shape parameter β are listed in Table 3. α indicates the breakdown strength and the PD resistance time when the failure probability is 63.2%; β indicates the dispersibility of the experimental data.

Table 3. Weibull parameters of DC breakdown and PD resistance.

Specimen	Weibull parameters			
	DC breakdown strength		PD resistance time	
	α (kV/mm)	β	α (s)	β
Pure PP	235.9	10.6	981	7.5
PP/Al ₂ O ₃ -1wt.%	322.5	12.8	1146	9.8
PP/Al ₂ O ₃ -7wt.%	213.8	12.6	3094	12.7
PP/Al ₂ O ₃ -15wt.%	186.2	11.6	3728	16.6
PP/Al ₂ O ₃ -20wt.%	168.3	9.4	4191	25.3

From the Weibull parameters, it is shown that the breakdown strength of PP/Al₂O₃-1wt% is 36.7% higher than that of pure PP, whereas that for 7, 15 and 20 wt% NCs are decreased by 9.4, 21.1 and 28.6%, respectively. In comparison, the PD resistance time increases with an increase of nanofiller content.

3.5 ERODED SURFACE MORPHOLOGY AFTER PD

The eroded surface morphology after PD discharge and its corresponding enlarged figures are shown in Figure 8. For pure PP and PP/Al₂O₃-1wt%, a hole is observed but the surrounding area (outside the yellow circle) was not eroded.

On the other hand, for PP/Al₂O₃-7, 15 and 20wt% NCs, the eroded area appears more spreading on the sample surface. In addition, several distinguishable eroded layers surrounding the hole can be observed, as shown in Figures 8c and 8d. In the corresponding enlarged figures (2000 magnification), it is found that the polymeric matrix in the center layer is the most eroded. With increase of the distance to the center hole, the erosion degree on sample surface is gradually reduced.

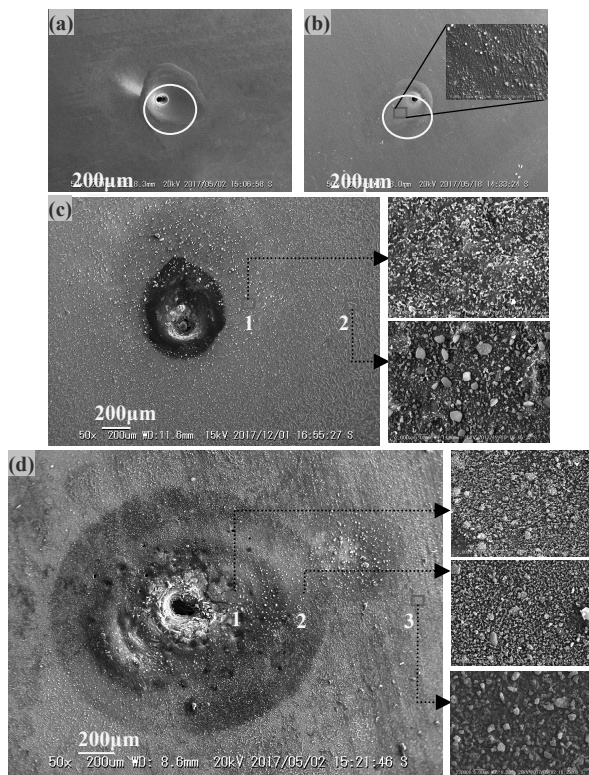


Figure 8. Eroded surface morphology of (a) pure PP, (b) NCs of 1 wt%, (c) 7 wt%, (d) 15 and 20 wt%.

4 DISCUSSION

4.1 DC BREAKDOWN RELATED TO TRAP-CONTROLLED CHARGE TRANSPORT

Space charge transport is closely related to dielectric breakdown [19], which includes charge migration, trapping, de-trapping, and recombination processes. Moreover, charge transport is affected by trap characteristics [3–7]. The interface region is the bridge that connecting the nanoparticles with the polymer matrix. The charge transport process is dominated by the role of interface that plays an important part on the traps [17–21]. In our previous work, a model for the interface region around the surface-treated nanoparticle was proposed [6, 20], which includes the bonded and the transition regions, as shown in Figure 9a. The model speculates that a vast distortion of potential occurs in the bonded region and deeper

traps are introduced due to abundance of unsaturated bonds such as hydrogen and organic groups (coupling agent). These bonds exist on the surface of nanoparticles, acting as deep traps in capturing carriers and restraining the carrier migration. Charges are diffused in the Debye shielding length, thus creating a local electrical field and repulsing other charge carriers, in which charges decay in exponentially with distance according to the Born approximation. Traps are distributed from deep to shallow outwards from the bonded region owing to the reduced coulomb potential [4–6].

With an increase in nanofiller content, the interfacial volume fraction increases and the average distance between neighboring nanoparticles becomes shorter. Some interface regions overlap with each other [5, 20], as illustrated in Figure 9b. Fr chet te reported that large filler concentration (20 wt%) resulted in greater interphase overlapping between the nanoparticles [22], which was consistent with the proposed interface model at high nanofillers contents[20]. In this case, the shallow traps gradually take a dominating role due to the large proportion of transition regions [5–6]. Percolated conductive paths are thus easily formed through the overlapped transition regions in which large quantities of shallow traps reside [7, 20, 23]. In addition, according to the Q_{TSDC} and conductivity results, an increase in conductivity and density of shallow traps were observed for NCs with high content that is consistent with the analysis above.

Charge dynamics include charge transport, energy gain and dissipation, impact ionization, charge multiplication, etc. [17–19]. The impact ionization is affected by carrier mobility and energy [17]. Additionally, carrier migration is closely related to traps. Traps would affect the mobility and kinetic energy due to the trapping and de-trapping process [24]. The schematic of electrons traps distribution in the energy band [17, 19] of polymeric material and trap-controlled charge transport process are shown in Figure 10. The deeper trap has greater trapping effect [17, 18] on the carriers. It is difficult for the trapped carriers in deeper traps to escape, thus inhibiting carrier migration and decreasing carrier energy. This leads to decreased carrier mobility as quantified in Equation (4) [25], finally leading to the mitigation of initial impact ionization and the improvement of breakdown performance.

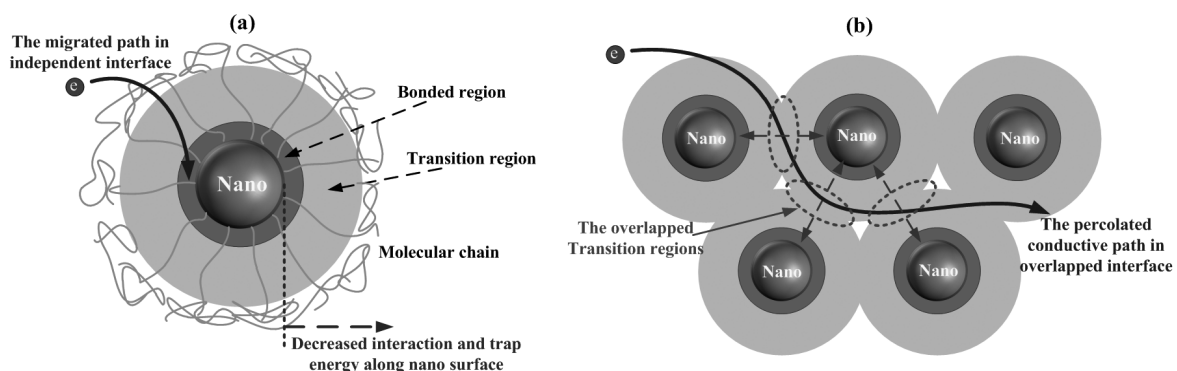


Figure 9. The carrier migrated characteristics induced by different nanofiller contents, (a) independent interface region around nanoparticles at low nanofiller content, (b) overlapped interface regions at high nanofiller contents.

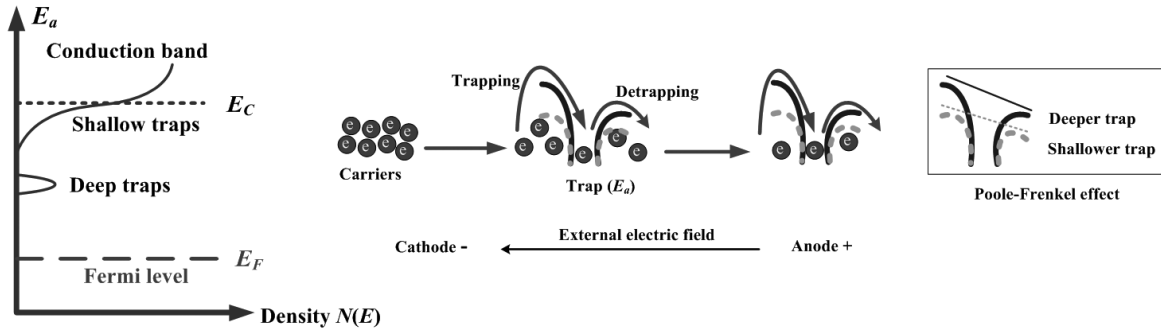


Figure 10. Schematic of electrons traps distribution in energy band of polymeric material and DC breakdown related to trap-controlled carriers transport.

Conversely, shallower traps facilitate carrier migration. This makes it easier to induce impact ionization and charge multiplication, finally decreasing the breakdown strength. DC breakdown strength related to trap energy is quantified as Equation (5) [18]. The shallower the trap energy is, the lower the breakdown strength becomes. The effective carrier mobility μ_{eff} and the critical breakdown strength F_c related to trap energy are given by:

$$\mu_{eff} = \mu_0 \exp\left(-\frac{E_a}{k_B T}\right) \quad (4)$$

$$F_c \propto \exp\left(\frac{E_a}{4k_B T}\right) \quad (5)$$

where μ_0 is the electronic mobility at the edge of the conduction band, and E_a is the trap energy.

To verify the above analysis more directly, the relationships among trap energy, effective carrier mobility, and DC breakdown strength are summarized in Figure 11. It is shown that the deeper the trap is, the lower the carrier mobility with higher breakdown strength becomes. The trend of trap energy is in accordance with the DC breakdown strength at different filler contents. Specifically, the trap energy of PP/Al₂O₃-1wt% is increased compared with that of pure PP, and the enhanced trapping effect can greatly restrict carrier migration, leading to the decrease of carrier mobility and conductivity. In contrast, for PP/Al₂O₃-7, 15 and 20 wt% possess more overlapped interfaces and shallower traps, which facilitate carrier migration and multiplication, thus easier leading to breakdown.

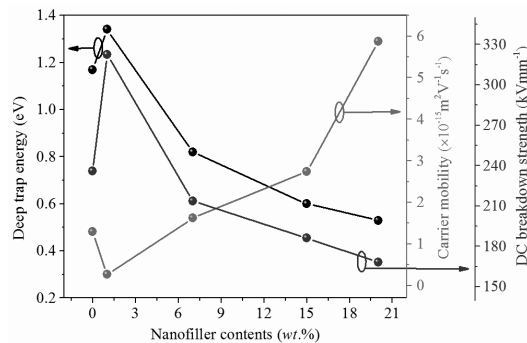


Figure 11. Trap energy, carrier mobility, and DC breakdown strength change with different nanofiller contents.

4.2 PD RESISTANCE RELATED TO ENERGY DISSIPATION

Owing to the extremely uneven electric field during the discharge process, electrons emitted from needle electrode interact with gaseous (air) molecules in the needle-sample gap, negative and positive ions are produced under the needle electrode by impact ionization, thermal ionization, and attachment effect [13]. The electrons, positive and negative ions are collectively referred to as charged ions. Surface degradation is caused by the energetic charged ions that directly collide with the polymer molecules [10–13].

For pure PP and PP/Al₂O₃-1wt%, the eroded surface morphology is a hollow pit. During the discharge erosion on sample surface, the energy of the charged ions is directly transferred to the PP molecules, causing fragmentation of the molecular chains. The area below the needle tip suffers the most severe degradation. The PD resistance of PP/Al₂O₃-1wt% is better than that of pure PP. This could be attributed to the blocking effect by the inorganic nanoparticles in the NCs which extends the migration path into a zigzag path [11, 14].

In this work, PD resistance model with the synergistic effects of energy dissipation by nano-scattering and charge dissipation is proposed for NCs with large amount of nanofillers, as shown in Figure 12. For PP/Al₂O₃-7, 15 and 20 wt% NCs, the PD resistance performance is obviously improved. Furthermore, the different eroded surface morphologies and the layered rings occurred in NCs inspire further exploration. Owing to the violent discharge on sample surface [10], the electrostatic Coulomb force around the nanoparticles is considered ineffective in shielding from the

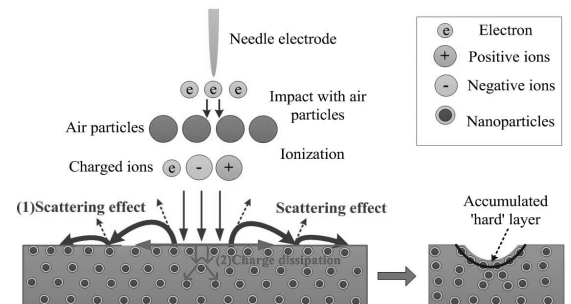


Figure 12. Synergistic effect of energy dissipation model by surface scattering effect and charge dissipation characteristics.

bombardment by the continuously energetic charged ions. The inorganic nanofillers are more discharge resistant compared to organic molecules because of its stronger bond energy and higher melting point [11, 13]. The organic molecules gradually degrade, resulting in accumulation of residual nanoparticles layer (the black line shown in Figure 12) on sample surface. The formation of such a 'hard' layer can be effective in blocking the erosion by subsequent energetic charged ions.

In Figure 13a, it is shown that the PD resistance time increases with dielectric permittivity (@ 50 Hz). When PD begins, the energetic charged ions are more concentrated towards the inorganic nanoparticles because the electric lines of force become denser around inorganic fillers with higher permittivity than around the organic matrix with lower permittivity [10–12]. As a result, there are more collisions between the charged ions and the inorganic fillers. Due to nanoparticle scattering effect arising from the 'hard' collision, the charged ions are able to be scattered to the outer area and undergo multiple collisions with the sample surface, as illustrated in Figure 12, resulting in the distinct layered rings on NCs samples with high filler content. Naturally, the total energy transferred from the charged ions to polymer molecules in a concentrated area is reduced owing to the energy dissipation by nano-scattering originated from the 'hard' layer. The area below the needle tip where suffers the most severe erosion is therefore protected. Besides, the surface nano-scattering effect is more prominent for NCs at higher nanofiller content, thus leading to the better PD resistance performance.

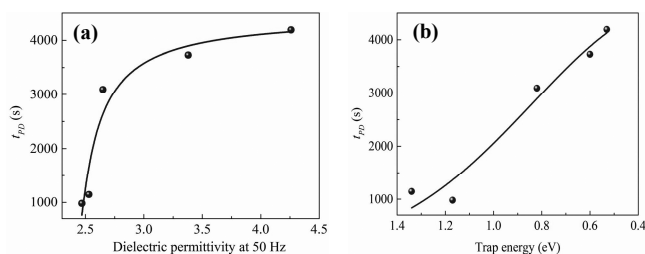


Figure 13. PD resistance time t_{PD} changed with (a) dielectric permittivity at 50 Hz and (b) trap energy.

Moreover, charged ions are mainly concentrated on the sample surface under alternating applied electric field. Hence, information about the surface charge dissipation should be considered. In Figure 13b, it is shown that the PD resistant time increases along with the decreased trap energy. The shallower traps introduced by incorporating large amount of nanofillers, as shown in TSDC result, facilitate charge dissipation along the sample surface. As a result, less energy of the charged ions is transferred to the polymer molecules during the trapping and de-trapping processes by exchanging energy [13]. Hence, the total transferred energy is reduced owing to the faster charge dissipation originated from the shallower traps. Finally, the centralized erosion and energy concentration in the sample bulk are mitigated. In summary, nano-scattering effect, higher dielectric permittivity and

shallower traps are the factors that help to disperse the energy concentration and contribute to longer PD resistance time.

5 CONCLUSIONS

This paper investigates the DC breakdown and PD resistance performance of PP/Al₂O₃ NCs incorporated with different nanofiller contents. The main conclusions are summarized below:

(1) The effect of nanofiller contents on DC breakdown and PD resistance was different. The DC breakdown strength of PP/Al₂O₃-1wt% was 36.7% higher than that of pure PP, while that of PP/Al₂O₃-7, 15 and 20 wt% were successively decreased. Comparatively, the PD resistance time increased with an increase in nanofiller content, and more obvious at high nanofiller contents.

(2) DC breakdown was closely related to trap-controlled charge transport. The trap energy instigated by the interface region affects charge transport. The deeper trap energy has greater trapping effect on restricting charge carriers, thus decreasing the carrier mobility and energy, leading to the lag occurs of impact ionization and the improvement of breakdown strength.

(3) PD resistance model based on synergistic effects of energy dissipation by nanoparticle scattering and charge dissipation was analyzed. Nano-scattering effect, which colliding against the charged ions played an important role in energy dissipation. Faster charge dissipation along the specimen surface induced by shallower traps mitigated energy concentration. Two factors contribute to better PD resistance and interpret the occurrence of layered rings for NCs with high filler contents.

ACKNOWLEDGMENT

This work is supported by the National Basic Research Program of China (973 Program) under Project No. 2015CB251003, and the National Natural Science Foundation of China (NSFC) under Project No. 51337008.

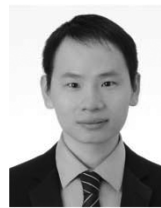
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